



1.6 PRACTICAL EXAMPLES WITH POTENTIOMETERS

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As previously stated, potentiometers are most commonly used in amps, radio and TV receivers, cassette players and similar devices. They are used for adjusting volume, tone, balance, etc.

As an example, we will analyze the common circuit for tone regulation in an audio amp. It contains two pots and is shown in the figure 1.8a.

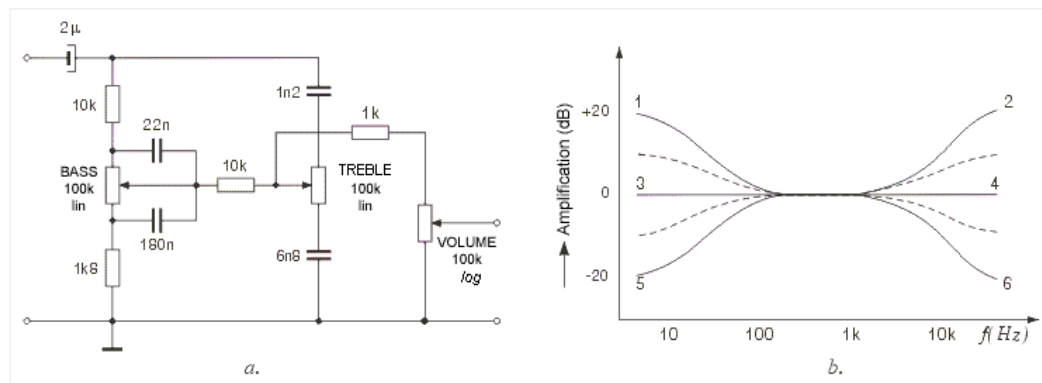


Fig. 1.8 Tone regulation circuit: a. Electrical scheme, b. Function of amplification

Potentiometer marked BASS regulates low frequency amplification. When the slider is in the lowest position, amplification of very low frequency signals (tens of Hz) is about ten times greater than the amplification of mid frequency signals ($\sim$ kHz). If slider is in the uppermost position, amplification of very low frequency signals is about ten times lower than the amplification of mid frequency signals. Low frequency boost is useful when listening to music with a beat (disco, jazz, R&B...), while Low Frequency amplification should be reduced when listening to speech or classical music.

Similarly, potentiometer marked TREBLE regulates high frequency amplification. High frequency boost is useful when music consists of high-pitched tones such as chimes, while for example High Frequency amplification should be reduced when listening to an old record to reduce the background noise.

Diagram 1.8b shows the function of amplification depending upon the signal frequency. If both sliders are in their uppermost position, the result is shown with curve 1-2. If both are in mid position function is described with line 3-4, and with both sliders in the lowest position, the result is shown with curve 5-6. Setting the pair of sliders to any other possible results in curves between curves 1-2 and 5-6.

Potentiometers BASS and TREBLE are coated by construction and linear by resistance.

The third pot in the diagram is a volume control. It is coated and logarithmic by resistance (hence the mark *log*)



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